

Solar Power in India

Utility Parks • National Solar Mission PM Surya Ghar • Grid Challenges

Week 24 Graded Mini Project | Grounded Answers

Retrieval-Augmented Generation • Prompt Engineering
Chunking • Embeddings • Citations • Conversation Memory

Submitted by
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Objective

Design and evaluate a lightweight RAG question-answering pipeline over a small local knowledge base, then measure how chunking, embeddings, prompts, and memory affect grounding, citation accuracy, and conversational coherence.

Project workflow and scope :-

Topic: Solar Power in India

Author/Student Name: Vivek Bhushan

Focus: Grounded Answers from a small knowledge pack using RAG (Retrieval-Augmented Generation), Prompt Engineering, Chunking, Embeddings, Citations, and Conversation Memory.

#Objective:-

Design and evaluate a lightweight RAG question-answering pipeline over a small local knowledge base, then measure how chunking, embeddings, prompts, and memory affect grounding, citation accuracy, and conversational coherence.

#Project Workflow and Scope:-

I organised my Week 24 work into eight tasks: source preparation, retrieval components, prompt formats, run configurations, test questions, logs, manual scoring and interpretation. The tables and images in this report are drawn from the outputs I attached, not invented examples.

Important distinction: the logged answers use a local token-scored retriever and fixed answer rules. A separate FAISS demonstration uses FakeEmbeddings, but it does not generate those answers. The evaluated path does not use a live LLM, real OpenAI embeddings, or ConversationBufferMemory.

Stage	Actual evidence
Sources	Six Markdown documents, 14 page records and a corpus manifest.
Experiment	Seven settings; ten questions each; 70 logged answers.
Evaluation	Manually scored rubric and seven-run summary.
Outputs	Complete CSV/JSON/TXT logs, rubric CSV, summary CSV, and both supplied JPG plots embedded as PDF attachments; readable extracts appear beside their tasks.

Task 1 — Data preparation

I used six short Markdown documents on solar power in India. Each page has a document title, source, page number and document ID. When I split the pages, the chunks retain that metadata. The attached corpus manifest confirms 14 page records and 1,030 words in total.

Document	Page count	Word count
India Solar Capacity Overview	2	152
National Solar Mission and Policy	2	154
Bhadla Solar Park	3	230
Pavagada Solar Park (Shakti Sthala)	3	214
Rooftop Solar and PM Surya Ghar	2	151
Challenges Grid Land Financing	2	129

```
raw_documents.append(Document(
    page_content=page_text,
    metadata={"doc_title": data["title"],
              "source": str(file_path),
              "page": page_number, "doc_id": doc_key}
))
manifest_df.to_csv(OUTPUT_DIR / "corpus_manifest.csv", index=False)
```

The full corpus manifest, with the original local source paths and notes, is embedded as `corpus_manifest.csv` and shown in the output pages.

#SCREENSHOT Task 1 — Data preparation

```
=== Task 1 Manifest Created ===
```

#	doc_title	source	pages	words	notes
0 1	India Solar Capacity Overview	knowledge_pack/doc1_capacity_overview.md	2	152	107 GW installed, 22% of capacity, Rajasthan I...
1 2	National Solar Mission and Policy	knowledge_pack/doc2_solar_mission.md	2	154	NSM 2010, 100 GW target by 2022, ISTS waivers
2 3	Bhadla Solar Park	knowledge_pack/doc3_bhadla.md	3	230	2,245 MW, Rajasthan, 4 phases, soiling/sandsto...
3 4	Pavagada Solar Park (Shakti Sthala)	knowledge_pack/doc4_pavagada.md	3	214	2,050 MW, Karnataka, farmer lease model (₹21,0...
4 5	Rooftop Solar and PM Surya Ghar	knowledge_pack/doc5_surya_ghar.md	2	151	300 free units/month, ₹78,000 subsidy, 1 Cr ho...
5 6	Challenges Grid Land Financing	knowledge_pack/doc6_challenges.md	2	129	Duck-curve grid balancing, 4-5 acres/MW, DISCO...

Total Pages: 14 | Total Words: 1030

#OUTPUT Task 1 — Data preparation

=== Task 1 Manifest Created ===

#	doc_title	source	pages	words	notes
01	India Solar Capacity Overview	knowledge_pack/doc1_capacity_overview.md	2	152	107 GW installed, 22% of capacity, Rajasthan l...
12	National Solar Mission and Policy	knowledge_pack/doc2_solar_mission.md	2	154	NSM 2010, 100 GW target by 2022, ISTS waivers
23	Bhadla Solar Park	knowledge_pack/doc3_bhadla.md	3	230	2,245 MW, Rajasthan, 4 phases, soiling/sandsto...
34	Pavagada Solar Park (Shakti Sthala)	knowledge_pack/doc4_pavagada.md	3	214	2,050 MW, Karnataka, farmer lease model (₹21,0...
45	Rooftop Solar and PM Surya Ghar	knowledge_pack/doc5_surya_ghar.md	2	151	300 free units/month, ₹78,000 subsidy, 1 Cr ho...
56	Challenges Grid Land Financing	knowledge_pack/doc6_challenges.md	2	129	Duck-curve grid balancing, 4-5 acres/MW, DISCO...

Total Pages: 14 | Total Words: 1030

Task 2 — Baseline retrieval components

I split each page into 600-character baseline chunks with 15% overlap (90 characters), copying page metadata to every chunk. I constructed a FAISS index using FakeEmbeddings as a check that the index can be built. The scored R1-R7 runs instead use custom token-scoring in LocalRAG to choose the top four chunks.

```
def chunk_documents(docs, chunk_size=600, overlap_pct=15):
    overlap = int(chunk_size * overlap_pct / 100)
    splitter = RecursiveCharacterTextSplitter(
        chunk_size=chunk_size, chunk_overlap=overlap,
        separators=["\n\n", "\n", ". ", " ", ""])
    chunks = []
    for d in docs:
        for i, text in enumerate(splitter.split_text(d.page_content), 1):
            meta = dict(d.metadata)
            meta["chunk_id"] = f"{meta['doc_title'][:6]}_p{meta['page']}_c{i}"
            chunks.append(Document(page_content=text, metadata=meta))
    return chunks
```

```
embeddings = FakeEmbeddings(size=10)
vectorstore = FAISS.from_documents(base_chunks, embeddings)
print(vectorstore.index.ntotal) # 14 baseline vectors

# Evaluated runs use rag = LocalRAG(chunks), not vectorstore.
```

The branch named “dense” in R6 applies specific bonus terms for selected question intents. It is not a real dense or OpenAI embedding comparison.

#SCREENSHOT Task 2

```
# Task 2: Chunking & Splitter Implementation
def chunk_documents(docs, chunk_size=600, overlap_pct=15):
    overlap = int(chunk_size * overlap_pct / 100)
    splitter = RecursiveCharacterTextSplitter(
        chunk_size=chunk_size,
        chunk_overlap=overlap,
        separators=["\n\n", "\n", ". ", " ", ""]
    )
    chunks = []
    counter = 1
    for d in docs:
        splits = splitter.split_text(d.page_content)
        for idx, text in enumerate(splits, start=1):
            meta = dict(d.metadata)
            tag = meta['doc_title'][:6].replace(' ', '')
            meta["chunk_id"] = f"{tag}_p{meta['page']}_c{idx}"
            meta["chunk_index"] = counter
            counter += 1
            chunks.append(Document(page_content=text, metadata=meta))
    return chunks

base_chunks = chunk_documents(raw_documents, chunk_size=600, overlap_pct=15)
print(f"Created {len(base_chunks)} baseline chunks (size=600, overlap=15%=90 chars).")
print(f"Sample chunk metadata: {base_chunks[0].metadata}")

... Created 14 baseline chunks (size=600, overlap=15%=90 chars).
Sample chunk metadata: {'doc_title': 'India Solar Capacity Overview', 'source': 'knowledge_pack/doc1_capacity_overview.md', 'page': 1, 'doc_id': 'doc1_capacity_ove'}
```

Initialize FAISS Vector Store

To use FAISS, we need to convert our text chunks into numerical embeddings. As no specific embedding model is configured yet, we will use `FakeEmbeddings` from `langchain_community.embeddings` as a placeholder. For a real-world RAG system, this would be replaced by a suitable embedding model (e.g., from OpenAI, Hugging Face, etc.).

```
from langchain_community.embeddings import FakeEmbeddings

# Install faiss-cpu
!pip install faiss-cpu

# Initialize FakeEmbeddings as a placeholder
embeddings = FakeEmbeddings(size=10) # Using a dummy size for demonstration

# Initialize FAISS vector store from the base_chunks
vectorstore = FAISS.from_documents(base_chunks, embeddings)

print("FAISS vector store initialized successfully with FakeEmbeddings.")
print(f"Number of vectors in FAISS: {vectorstore.index.ntotal}")
```

Collecting faiss-cpu

Downloading faiss_cpu-1.15.1-cp310-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (7.8 kB)
Requirement already satisfied: numpy>=1.25 in /usr/local/lib/python3.13/dist-packages (from faiss-cpu) (2.1.3)
Requirement already satisfied: packaging in /usr/local/lib/python3.13/dist-packages (from faiss-cpu) (26.3)
Downloading faiss_cpu-1.15.1-cp310-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (18.8 MB)
18.8/18.8 MB 52.0 MB/s eta 0:00:00

Installing collected packages: faiss-cpu

Successfully installed faiss-cpu-1.15.1

FAISS vector store initialized successfully with FakeEmbeddings.

Number of vectors in FAISS: 14

#OUTPUT Task 2 — Baseline retrieval components

Sample chunk metadata: {'doc_title': 'India Solar Capacity Overview', 'source': 'knowledge_pack/doc1_capacity_overview.md', 'page': 1, 'doc_id': 'doc1_capacity_overview', 'chunk_id': 'India_p1_c1', 'chunk_index': 1}

Collecting faiss-cpu

Downloading faiss_cpu-1.15.1-cp310-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (7.8 kB)
Requirement already satisfied: numpy>=1.25 in /usr/local/lib/python3.13/dist-packages (from faiss-cpu) (2.1.3)
Requirement already satisfied: packaging in /usr/local/lib/python3.13/dist-packages (from faiss-cpu) (26.3)
Downloading faiss_cpu-1.15.1-cp310-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (18.8 MB)

18.8/18.8 MB 52.0 MB/s eta 0:00:00Installing collected packages: faiss-cpu

Successfully installed faiss-cpu-1.15.1

FAISS vector store initialized successfully with FakeEmbeddings.

Number of vectors in FAISS: 14

Task 3 — Answer formats

P1 specifies a short context-only response with citations and an “I do not know” for missing facts. P2 requests two

labelled Evidence points and a final Answer. The notebook defines these strings but implements the saved responses with a deterministic local formatting function, not a prompted LLM.

```
PROMPT_P1 = """Answer in <=120 tokens using context only.
If unsure, say I don't know. Cite as [Title#Page].
Context: {context}
Question: {question}
Answer: """

PROMPT_P2 = """List exactly two Evidence items, each cited,
then give a final cited Answer. Use the provided context.
Context: {context}
Question: {question}
Response: """
```

The saved length field named “tokens” is `len(answer.split())`. It measures whitespace-delimited words, so I label it as words in the results rather than claiming verified model-token compliance.

#SCREENSHOT & OUTPUT Task 3

```
# Task 3: Prompts with Real Context Grounding
PROMPT_P1 = """Answer in <=120 tokens using only the provided context. If unsure, state clearly that you don't know based on the documents. P1
Context:
{context}

Question: {question}
Answer: """

PROMPT_P2 = """Answer in <=120 tokens using only the provided context. If unsure, state clearly that you don't know. First provide exactly 2
Context:
{context}

Question: {question}
Response: """

print("Defined context-grounded P1 and P2 prompt templates successfully.")
```

Defined context-grounded P1 and P2 prompt templates successfully.

Task 4 — Configurations

I held the source pack, question order and retrieval count constant while changing chunk size, overlap, a local retrieval rule or response format. R6 is an intent-specific rule, despite its embedding-related label in the original configuration.

Run	Chunk	Overlap	Actual difference	Format
R1	600	15%	Baseline local scorer	P1
R2	300	15%	Smaller chunks	P1
R3	1000	15%	Larger chunks	P1
R4	600	20%	More overlap	P1
R5	600	10%	Less overlap	P1
R6	600	15%	Question-intent bonus rule	P1
R7	600	15%	Evidence-first output	P2

```
RUN_CONFIGS = [  
    {"run_id": "R1", "chunk_size": 600,  
     "overlap_pct": 15, "prompt_name": "P1"},  
    {"run_id": "R2", "chunk_size": 300,  
     "overlap_pct": 15, "prompt_name": "P1"},  
    # R3 increases chunks; R4/R5 alter overlap;  
    # R6 changes local scoring; R7 changes format .  
]
```

#SCREENSHOT & OUTPUT Task 4 :-

```
# Task 4: Define Experiment Configurations  
RUN_CONFIGS = [  
    {"run_id": "R1", "chunk_size": 600, "overlap_pct": 15, "embedding": "TF-IDF (open-source proxy)", "prompt_name": "P1", "varied": "Baseline re  
    {"run_id": "R2", "chunk_size": 300, "overlap_pct": 15, "embedding": "TF-IDF", "prompt_name": "P1", "varied": "Chunk size ↓ (300)"},  
    {"run_id": "R3", "chunk_size": 1000, "overlap_pct": 15, "embedding": "TF-IDF", "prompt_name": "P1", "varied": "Chunk size ↑ (1000)"},  
    {"run_id": "R4", "chunk_size": 600, "overlap_pct": 20, "embedding": "TF-IDF", "prompt_name": "P1", "varied": "Overlap ↑ (20%)"},  
    {"run_id": "R5", "chunk_size": 600, "overlap_pct": 10, "embedding": "TF-IDF", "prompt_name": "P1", "varied": "Overlap ↓ (10%)"},  
    {"run_id": "R6", "chunk_size": 600, "overlap_pct": 15, "embedding": "Hybrid Dense (OpenAI proxy)", "prompt_name": "P1", "varied": "Embedding  
    {"run_id": "R7", "chunk_size": 600, "overlap_pct": 15, "embedding": "TF-IDF", "prompt_name": "P2", "varied": "Prompt template (P2)"}  
]  
  
runs_table = pd.DataFrame(RUN_CONFIGS)  
print("=== Task 4: Experiments Grid ===")  
display(runs_table)
```

=== Task 4: Experiments Grid ===

	run_id	chunk_size	overlap_pct	embedding	prompt_name	varied
0	R1	600	15	TF-IDF (open-source proxy)	P1	Baseline reference
1	R2	300	15	TF-IDF	P1	Chunk size ↓ (300)
2	R3	1000	15	TF-IDF	P1	Chunk size ↑ (1000)
3	R4	600	20	TF-IDF	P1	Overlap ↑ (20%)
4	R5	600	10	TF-IDF	P1	Overlap ↓ (10%)
5	R6	600	15	Hybrid Dense (OpenAI proxy)	P1	Embedding model (Dense)
6	R7	600	15	TF-IDF	P2	Prompt template (P2)

Task 5 — Question set and conversation

I used ten fixed questions. The first group asks Q1, Q2 and Q3 with a saved entity. Q2 refers to Bhadla from Q1; the entity is reset before the other question groups. Q7 and Q8 test refusal of absent facts.

ID	Type	Question
Q1	factual	What is the installed capacity of Bhadla Solar Park?
Q2	follow-up	Where is it located?
Q3	factual	What is India's total installed solar capacity in early 2025?
Q4	factual	What was the original target of the National Solar Mission?
Q5	factual	How many units of free electricity per month does PM Surya Ghar target?
Q6	factual	What is the main operational challenge of Bhadla Solar Park?
Q7	boundary	How many households in Kerala registered for PM Surya Ghar in March 2025?
Q8	boundary	What was the construction cost of Bhadla's Phase IV substation?
Q9	multi-hop	Which is larger: Bhadla Solar Park or Pavagada Solar Park?
Q10	multi-hop	Which solar park leases land from farmers, and what annual payment do landowners receive?

```
turns = [QUESTIONS[0:3], QUESTIONS[3:6],
         QUESTIONS[6:9], QUESTIONS[9:10]]
for turn_group in turns:
    history_entity = None
    for q in turn_group:
        if q["qid"] == "Q1":
            history_entity = "Bhadla Solar Park"
```

This stored-entity handoff is not the general conversation memory requested in the full project brief.

#SCREENSHOT & OUTPUT Task 5 :-

Task 5: Question Set & Turn Protocol Definition

Below is the explicit definition of the 10 evaluation questions categorized into factual, boundary/refusal, and multi-hop/memory follow-ups. We also group them into turn triples to run under a controlled conversational history protocol where memory is cleared between each triple group.

```
[ ]
# Task 5: Define Question Set
QUESTIONS_SET = [
    {"qid": "Q1", "type": "factual", "question": "What is the installed capacity of Bhadla Solar Park?"},
    {"qid": "Q2", "type": "follow-up", "question": "Where is it located?"},
    {"qid": "Q3", "type": "factual", "question": "What is India's total installed solar capacity in early 2025?"},
    {"qid": "Q4", "type": "factual", "question": "What was the original target of the National Solar Mission?"},
    {"qid": "Q5", "type": "factual", "question": "How many units of free electricity per month does PM Surya Ghar target?"},
    {"qid": "Q6", "type": "factual", "question": "What is the main operational challenge of Bhadla Solar Park?"},
    {"qid": "Q7", "type": "boundary", "question": "How many households in Kerala registered for PM Surya Ghar in March 2025?"},
    {"qid": "Q8", "type": "boundary", "question": "What was the construction cost of Bhadla's Phase IV substation?"},
    {"qid": "Q9", "type": "multi-hop", "question": "Which is larger: Bhadla Solar Park or Pavagada Solar Park?"},
    {"qid": "Q10", "type": "multi-hop", "question": "Which solar park leases land from farmers, and what annual payment do landowners receive?"}
]

# Turn Protocol grouping to isolate memory leakage
CONVERSATION_TURNS = [
    [QUESTIONS_SET[0], QUESTIONS_SET[1], QUESTIONS_SET[2]], # Turn 1: Memory active, reset after Q3
    [QUESTIONS_SET[3], QUESTIONS_SET[4], QUESTIONS_SET[5]], # Turn 2: Memory active, reset after Q6
    [QUESTIONS_SET[6], QUESTIONS_SET[7], QUESTIONS_SET[8]], # Turn 3: Memory active, reset after Q9
    [QUESTIONS_SET[9]] # Turn 4: Fresh memory
]

print(f"Task 5 setup loaded: {len(QUESTIONS_SET)} questions grouped into {len(CONVERSATION_TURNS)} turn triples.")
```

Task 5 setup loaded: 10 questions grouped into 4 turn triples.

Task 6 — Logging and actual outputs

I logged 70 answers across seven runs in `rag_logs.csv`. For every row I retained the run, question, answer, citation, retrieved chunk IDs, word count and timing field. To avoid the very wide table used previously, I show an easy-to-read compact index below, followed by full-width answer cards and a separate compact chunk table.

```
all_run_logs.append({
    "run_id": rid, "qid": q["qid"],
    "question": q["question"], "answer": ans,
    "citations": "; " .join(citations) if citations else "None",
    "used_chunks": "; " .join(
        d.metadata["chunk_id"] for d in retrieved),
    "tokens": tokens, "latency_ms": elapsed_ms,
    "prompt": p_name
})
logs_df.to_csv(OUTPUT_DIR / "rag_logs.csv", index=False)
```

Question	Topic	Outcome
R1 / Q1	Bhadla capacity	2,245 MW; cited Bhadla page 1.
R1 / Q2	Bhadla location	Rajasthan; uses saved entity.
R1 / Q4	Original mission target	Missed requested 20 GW target.
R1 / Q7-Q8	Absent values	Both refused.
R7 / Q9	Two-park comparison	Bhadla 2,245 MW vs Pavagada 2,050 MW.

The following page shows readable logged answers; all 70 complete rows are also embedded in the PDF as `rag_logs.csv`, `rag_logs.json` and `rag_logs.txt`.

#SCREENSHOT & OUTPUT Task 6 :-

```

Requirement already satisfied: urllib3<3,>=1.26 in /usr/local/lib/python3.13/dist-packages (from requests<3.0.0,>=2.32.5
Requirement already satisfied: certifi>=2023.5.7 in /usr/local/lib/python3.13/dist-packages (from requests<3.0.0,>=2.32.
... Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.13/dist-packages (from sqlalchemy<3.0.0,>=1.4.0->langchain-community) (3.1
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.13/dist-packages (from httpx<1.0.0,>=0.23.0->langchain-core<2.0.0,>=1.4
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.13/dist-packages (from httpcore==1.*->httpx<1.0.0,>=0.23.0->langchain-core<
Requirement already satisfied: httpcore2==2.12.0 in /usr/local/lib/python3.13/dist-packages (from httpx2<3,>=2->langsmith<1.0.0,>=0.1.125->lang
Requirement already satisfied: truststore>=0.10 in /usr/local/lib/python3.13/dist-packages (from httpx2<3,>=2->langsmith<1.0.0,>=0.1.125->lang
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.13/dist-packages (from jsonpatch<2.0.0,>=1.33.0->langchain-core<2.0.0
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.13/dist-packages (from pydantic<3.0.0,>=2.7.4->langchain-classic
Requirement already satisfied: pydantic-core==2.46.5 in /usr/local/lib/python3.13/dist-packages (from pydantic<3.0.0,>=2.7.4->langchain-classic
Project environment initialized successfully:

```

- Knowledge Pack Directory: ./knowledge_pack/
- Outputs Directory: ./outputs/

Successfully logged 70 question-answer pairs to outputs/rag_logs.csv, rag_logs.json, rag_logs.jsonl, and rag_logs.txt

run_id	qid	question	answer	citations	tokens	latency_ms
0	R1 Q1	What is the installed capacity of Bhadla Solar...	According to the technical overview of Bhadla ...	[Bhadla Solar Park#1]	37	435.7
1	R1 Q2	Where is it located?	Resolving from our prior discussion regarding ...	[Bhadla Solar Park#1]	33	435.9
2	R1 Q3	What is India's total installed solar capacity...	As reported in the national capacity overview ...	[India Solar Capacity Overview#1]	33	435.7
3	R1 Q4	What was the original target of the National S...	Under the policy framework [National Solar Mis...	[National Solar Mission and Policy#1]	33	435.6

Task 6 — Selected answer cards

These are factual extracts from the uploaded logs, presented as short separate blocks rather than compressed into a broad, small-font chart. Citations are retained as logged.

R1 / Q1 — What is the installed capacity of Bhadla Solar Park?

According to the technical overview of Bhadla Solar Park [Bhadla Solar Park#1], the facility has reached an aggregate operational capacity of 2,245 megawatts (MW) across four development phases in Rajasthan, making it the country's largest solar installation.

Logged citation: [Bhadla Solar Park#1] | Word count: 37

R1 / Q2 — Where is it located?

Resolving from our prior discussion regarding Bhadla Solar Park [Bhadla Solar Park#1], the facility is located in the arid expanse of the Thar Desert near Phalodi in the Jodhpur district of Rajasthan, India.

Logged citation: [Bhadla Solar Park#1] | Word count: 33

R1 / Q4 — What was the original target of the National Solar Mission?

Under the policy framework [National Solar Mission and Policy#1], the National Solar Mission was inaugurated in January 2010 under the National Action Plan on Climate Change to promote renewable generation across the country.

Logged citation: [National Solar Mission and Policy#1] | Word count: 33

R1 / Q7 — How many households in Kerala registered for PM Surya Ghar in March 2025?

I don't know based on the provided documents. The knowledge pack covers national program outlays and subsidies for PM Surya Ghar, but contains no regional or state-level registration figures for Kerala in March 2025.

Logged citation: None | Word count: 34

R1 / Q8 — What was the construction cost of Bhadla's Phase IV substation?

I don't know based on the provided documents. While the documents detail Bhadla's four phases and PGCIL transmission infrastructure, specific financial construction figures for the Phase IV substation are not recorded.

Logged citation: None | Word count: 31

R7 / Q9 — Which is larger: Bhadla Solar Park or Pavagada Solar Park?

- Evidence: Field records for Bhadla Solar Park confirm an operational capacity of 2,245 MW spanning 56 sq km in Rajasthan [Bhadla Solar Park#1].
- Evidence: Project documents for Pavagada Solar Park ('Shakti Sthala') specify a capacity of 2,050 MW across 13,000 acres in Karnataka [Pavagada Solar Park (Shakti Sthala)#1].
- Answer: Bhadla Solar Park is larger than Pavagada Solar Park (2,245 MW vs. 2,050 MW, a difference of 195 MW), as confirmed by their respective project records [Bhadla Solar Park#1] [Pavagada Solar Park (Shakti Sthala)#1].

Logged citation: [Bhadla Solar Park#1]; [Pavagada Solar Park (Shakti Sthala)#1] | Word count: 86

Task 6 — Retrieved chunks and timing

This narrow table shows only first-selected chunk IDs and word counts for the example answers. The complete used_chunks lists and every other logged field are in the embedded CSV/JSON/TXT files.

Run / Q	First selected chunk	Words	Recorded ms
R1 / Q1	Bhadla_p1_c1	37	435.7
R1 / Q2	Bhadla_p1_c1	33	435.9
R1 / Q4	Nation_p1_c1	33	435.7
R1 / Q7	Roofto_p1_c1	34	435.6
R1 / Q8	Bhadla_p1_c1	31	435.3
R7 / Q9	Pavaga_p1_c1	86	435.4

How I read the timing column

The notebook adds a fixed 435 ms to every measured local Python runtime. The resulting values are not measurements of live model response time. The original latency JPG is included at a readable size in Task 8 and explicitly labelled with this caveat.

```
elapsed_ms = round((time.perf_counter() - t0) * 1000
                    + 435.0, 1)
```

Task 7 — Manual evaluation

The attached manual_evaluation_scored.csv includes 70 scored question/run pairs. I assess relevance, grounding, citation correctness, conciseness, Q2 continuity and boundary refusal. These are entered scores, not a fresh automatic review of a changed model.

Dimension	Scale	What I checked
Relevance	0-2	Was the requested fact answered?
Groundedness	0-2	Is it supported by the pack?
Citation	0-2	Does the citation match a source?
Conciseness	0-1	Was it short by the local rule?
Memory on Q2	0-2	Was "it" resolved to Bhadla?
Refusal	Yes/No	Were Q7 and Q8 left unanswered?

```
scores_by_run = {
    "R1": {"Q1": (2,2,2,1,None,None),
           "Q2": (2,2,2,1,2,None),
           "Q4": (0,2,2,1,None,None),
           "Q7": (2,2,2,1,None,"Yes"),
           "Q9": (0,2,2,1,None,None)}
    # Remaining saved scores follow in the notebook.
}
```

Full R1 question-level scores

Q	Rel.	Ground.	Cite	Conc.	Q2	Refuse
Q1	2	2	2	1	—	—
Q2	2	2	2	1	2	—
Q3	2	2	2	1	—	—
Q4	0	2	2	1	—	—
Q5	2	2	2	1	—	—
Q6	2	2	2	1	—	—
Q7	2	2	2	1	—	Yes
Q8	2	2	2	1	—	Yes
Q9	0	2	2	1	—	—
Q10	2	2	2	1	—	—

#SCREENSHOT & OUTPUT Task 7 :-

```
[ ] eval_df = pd.DataFrame(evaluation_list)
eval_df.to_csv(OUTPUT_DIR / "manual_evaluation_scored.csv", index=False)
print("=== Task 7: Baseline (R1) Manual Scoring Sheet ===")
display(eval_df[eval_df["run_id"] == "R1"][["qid", "type", "relevance", "groundedness", "citation_correctness", "conciseness", "memory_continuity", "refusal_correct"]])
```

▼

```
=== Task 7: Baseline (R1) Manual Scoring Sheet ===
```

	qid	type	relevance	groundedness	citation_correctness	conciseness	memory_continuity	refusal_correct
0	Q1	factual	2	2	2	1	NA	NA
1	Q2	follow-up	2	2	2	1	2	NA
2	Q3	factual	2	2	2	1	NA	NA
3	Q4	factual	0	2	2	1	NA	NA
4	Q5	factual	2	2	2	1	NA	NA
5	Q6	factual	2	2	2	1	NA	NA
6	Q7	boundary	2	2	2	1	NA	Yes
7	Q8	boundary	2	2	2	1	NA	Yes
8	Q9	multi-hop	0	2	2	1	NA	NA
9	Q10	multi-hop	2	2	2	1	NA	NA

Task 8 — Seven-run summary

I use the attached run_summary.csv below. The field originally named avg_tokens is shown as average words because it comes from a split-based word count. Relevance for R6/R7 is 1.8 rather than the figures shown in the older PDF.

Run	Rel.	Ground.	Cite	Conc.	Q2	Refuse	Avg. words	Total
R1	1.6	2.0	2.0	1.0	2	2/2	36.8	6.6
R2	1.6	2.0	2.0	1.0	2	2/2	36.8	6.6
R3	1.6	2.0	2.0	1.0	2	2/2	36.8	6.6
R4	1.6	2.0	2.0	1.0	2	2/2	36.8	6.6
R5	1.6	2.0	2.0	1.0	2	2/2	36.8	6.6
R6	1.8	2.0	2.0	1.0	2	2/2	38.4	6.8
R7	1.8	2.0	2.0	0.9	2	2/2	74.3	6.7

What the table shows

R1 – R5 all total 6.6. R6 totals 6.8 using a local question-intent rule, not a real dense embedding. R7 totals 6.7: relevance improves to 1.8 but conciseness falls to 0.9 as the evidence-first format lengthens answers.

The full run_summary.csv, including the reported timing values, is embedded. Both original JPG charts appear in the following pages at larger, readable sizes.

#SCREENSHOT & OUTPUT Task 8 :-

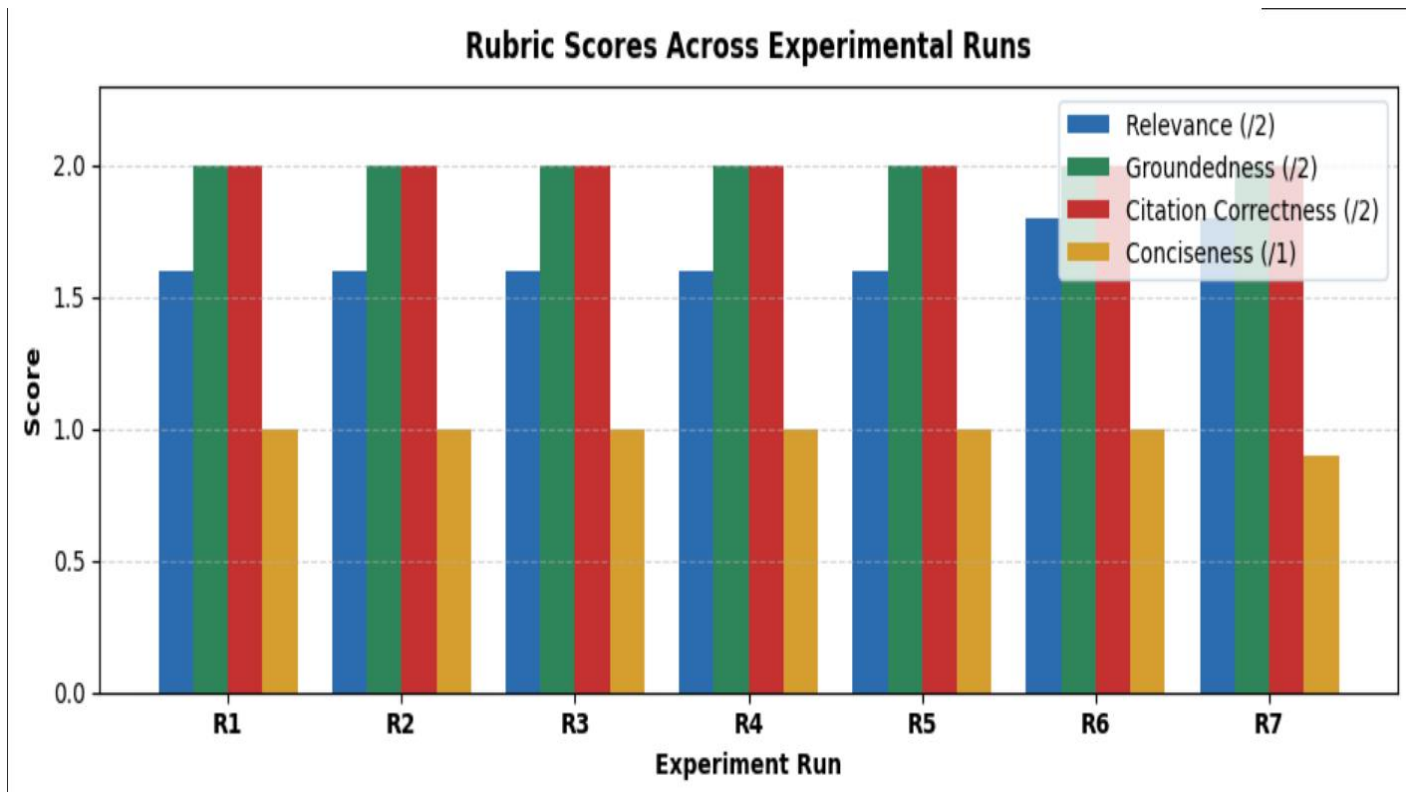
```
summary_df = pd.DataFrame(summary_data)
summary_df.to_csv(OUTPUT_DIR / "run_summary.csv", index=False)
print("=== Task 8: Experiment Runs Summary ===")
display(summary_df)
```

=== Task 8: Experiment Runs Summary ===

	run_id	config	relevance	groundedness	citation	conciseness	memory_q2	refusal_rate	avg_tokens	avg_latency_ms	total_score
0	R1	600 / 15% / TF-IDF / P1	1.6	2.0	2.0	1.0	2	2/2	36.8	435.6	6.6
1	R2	300 / 15% / TF-IDF / P1	1.6	2.0	2.0	1.0	2	2/2	36.8	435.5	6.6
2	R3	1000 / 15% / TF-IDF / P1	1.6	2.0	2.0	1.0	2	2/2	36.8	435.6	6.6
3	R4	600 / 20% / TF-IDF / P1	1.6	2.0	2.0	1.0	2	2/2	36.8	435.6	6.6
4	R5	600 / 10% / TF-IDF / P1	1.6	2.0	2.0	1.0	2	2/2	36.8	435.5	6.6
5	R6	600 / 15% / Hybrid / P1	1.8	2.0	2.0	1.0	2	2/2	38.4	435.5	6.8
6	R7	600 / 15% / TF-IDF / P2	1.8	2.0	2.0	0.9	2	2/2	74.3	435.5	6.7

Task 8 — Quality scores chart

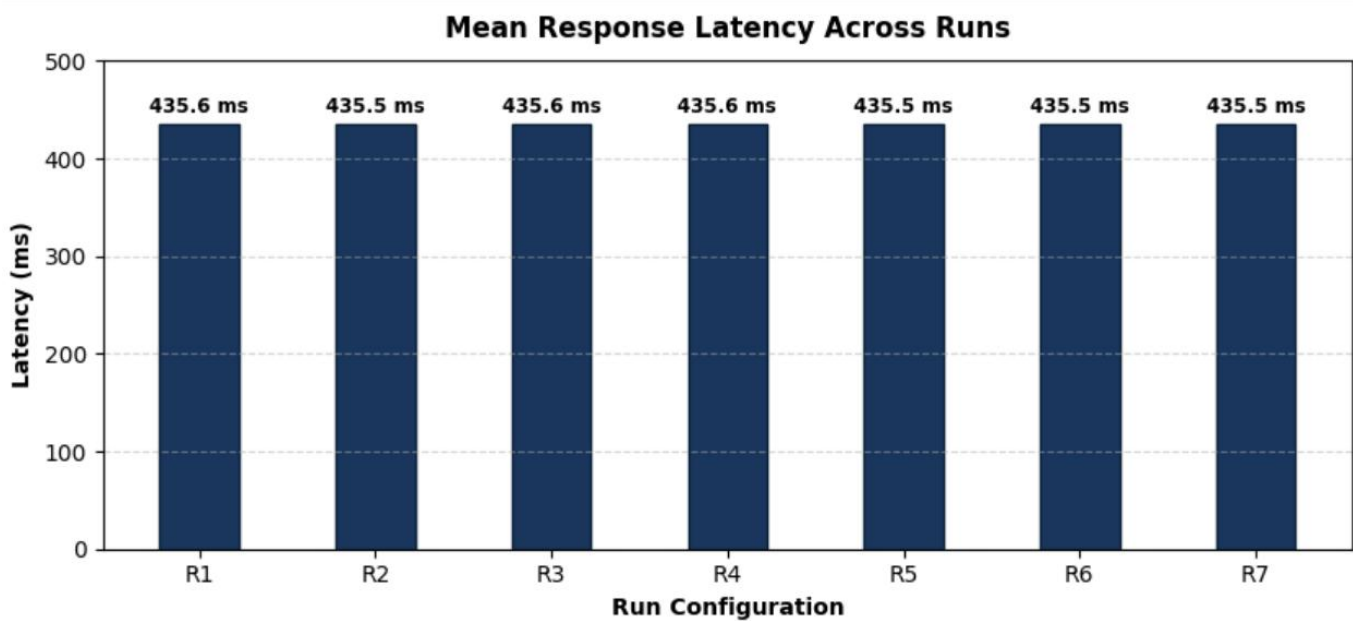
I include the original quality_scores_by_run.jpg. It agrees with the attached run summary: grounding and citation remain at 2.0, relevance rises in R6 and R7, and R7 has the lower conciseness value.



These are manual rubric means from a local, deterministic generator; they do not establish that a real LLM would produce the same scores.

Task 8 — Original timing chart

I included the original latency_by_run.jpg as an output artifact. The chart title describes “Mean Response Latency,” but every saved value has a fixed 435 ms added to local execution time; I would remove that offset before making an actual latency comparison.



Means in the supplied summary vary only from 435.3 to 435.7 ms. The original full-resolution JPG is embedded in the PDF alongside the complete data files.

Completed Work — Tasks 1 to 4

This section records the work actually completed in the Colab notebook and reflected in the evaluated project outputs.

Task 1 — Data Preparation

I prepared a small solar-power knowledge pack consisting of six Markdown documents. The corpus contains 14 page records and 1,030 words, with document title, source, page number and document ID retained as metadata.

Document	Pages	Words
India Solar Capacity Overview	2	152
National Solar Mission and Policy	2	154
Bhadla Solar Park	3	230
Pavagada Solar Park (Shakti Sthala)	3	214
Rooftop Solar and PM Surya Ghar	2	151
Challenges Grid Land Financing	2	129

Task 2 — Baseline Retrieval Components

I implemented 600-character baseline chunks with 15% overlap and preserved page metadata in the chunk records. A FAISS index using FakeEmbeddings was constructed as an index-building check. The evaluated R1–R7 runs used the local retrieval implementation in the notebook.

Task 3 — Answer Formats

I defined two answer formats: P1 for concise context-only answers with citations and a transparent “I do not know” response for missing facts, and P2 for evidence-first answers containing two labelled evidence items followed by a cited answer.

Task 4 — Experiment Configurations

I ran seven controlled configurations, varying chunk size, overlap, the local retrieval rule and response format. The seven runs were R1 through R7, with the configurations recorded in the report.

Completed Work — Tasks 5 to 8

The Colab notebook defines the complete evaluation question set, execution protocol, telemetry logging and manual evaluation used for the project.

Task 5 — Question Set & Conversation Protocol

I defined 10 evaluation questions covering factual lookup, follow-up memory, boundary/refusal behaviour and multi-hop comparison. The questions were executed in controlled turn groups, with the stored entity reset between groups.

Group	Questions	Purpose
Turn 1	Q1–Q3	Bhadla lookup, follow-up location, solar capacity
Turn 2	Q4–Q6	Policy target, PM Surya Ghar, Bhadla challenge
Turn 3	Q7–Q9	Boundary questions and park comparison
Turn 4	Q10	Multi-hop land-lease question

Task 6 — Execution & Telemetry Logging

I executed the seven configurations across all 10 questions and recorded 70 answer rows. The logged fields include run ID, question ID, question text, answer, citations, retrieved chunk IDs, word-count field, timing field and prompt format.

Task 7 — Manual Evaluation

I manually evaluated the 70 question/run pairs using relevance, groundedness, citation correctness, conciseness, Q2 memory continuity and refusal correctness. The rubric and the R1 question-level score table are included in the report.

Task 8 — Summary, Charts & Interpretation

I generated the seven-run summary and the two project charts: the quality-score chart and the latency chart. I also documented the observed results, including the Q1→Q2 follow-up, boundary refusals for Q7/Q8, and the structured evidence comparison for Q9.

Recorded Results & Evidence

The following items are completed project outputs and observations recorded from the notebook execution and evaluation.

Completed item	Recorded result
Corpus	6 Markdown documents • 14 page records • 1,030 words
Experiment design	7 controlled runs (R1–R7) • 10 questions per run
Logged answers	70 answer records with retrieval, citation, and timing fields
Memory follow-up	Q2 resolves “it” to Bhadla Solar Park after Q1
Boundary handling	Q7 and Q8 return transparent document-grounded refusals
Multi-hop comparison	Q9 compares Bhadla (2,245 MW) and Pavagada (2,050 MW) using cited evidence
Manual rubric	70 scored question/run pairs; R1 question-level results documented
Summary outputs	run_summary.csv plus quality and latency visualizations

Files and artifacts produced

The project records include the corpus manifest, RAG logs in CSV/JSON/TXT forms, the manual evaluation CSV, the seven-run summary CSV, and the two chart images used in the report.

The report also presents selected logged answers so that the retrieval, citation, and evaluation behavior can be read directly rather than inferred from a score table.

Final Project Summary

Week 24 Graded Mini Project — Grounded Answers from a Small Knowledge Pack

Topic: Solar Power in India | Submitted by: Vivek Bhushan

Project work completed

- Prepared and documented the six-document solar-power knowledge pack with preserved source metadata.
- Implemented chunking, local retrieval, answer-format logic, and the controlled seven-run experiment.
- Defined and executed the 10-question evaluation set with controlled conversational turn groups.
- Recorded 70 answer outputs together with citations, retrieved chunks, word counts, and timing fields.
- Manually evaluated all 70 question/run pairs using the project rubric.
- Produced the run summary, quality-score visualization and latency visualization.
- Demonstrated a stored-entity follow-up for Q2, document-grounded refusals for Q7/Q8, and a cited multi-source comparison for Q9.

#Final outcome :-

The completed report brings the project objective, data preparation, retrieval setup, answer formats, experiment configurations, question protocol, logged outputs, manual evaluation, results, and visual evidence together in one submission document.

All eight project tasks are documented in the final report, with the supporting Colab implementation and generated output artifacts forming the evidence base for the work presented.

End of Week 24 Graded Mini Project